



Department of Information Technology (Bsc.IT Semester III)
Python Programming Practical

Practical – 10

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a) Implement linear regression using breeze. Fit a model to a small dataset and predict a value

Code:

```
import breeze.linalg._
import breeze.stats._
import breeze.numetrics._ // 1. Fixed typo: 'numetrics' instead of 'numeric'

object Practen extends App {
  val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0)
  val y = DenseVector(6.0, 7.0, 8.0, 9.0, 10.0)

  val mX = mean(x)
  val mY = mean(y)

  val n = sum((x - mX) * (y - mY))
  val d = sum((x - mX) * (x - mX))

  val s = n / d
  val i = mY - s * mX

  println(f"Slope : $s%.2f")
  println(f"Intercept : $i%.2f")
}
```

```
val nx = 6.0  
val p = s * nx + i  
println(f"Predicted Value = $p%.2f")  
}
```

Output:

```
Slope : 1.00  
Intercept : 5.00  
Predicted Value = 11.00  
  
Process finished with exit code 0
```